Q1 2021 (31 Aug Shift 1)

A vertical pole fixed to the horizontal ground is divided in the ratio $3:7$ by a mark on it with lower part shorter than the upper part. If the two parts subtend equal angles at a point on the ground 18 m away from the base of the pole, then the height of the pole (in meters) is :

(1) $12\sqrt{15}$

(2) $12\sqrt{10}$

(3) $8\sqrt{10}$

(4) $6\sqrt{10}$

Q2 2021 (27 Aug Shift 2)

Two poles, AB of length a metres and CD of length $\mathrm{a}+\mathrm{b}(\mathrm{b} \neq \mathrm{a})$ metres are erected at the same horizontal level with bases at B and D. If $BD = x$ and $\tan \lfloor \mathrm{ACB} = \frac{1}{2}$, then:

(1) $x^2 + 2(a+2b)x - b(a+b) = 0$

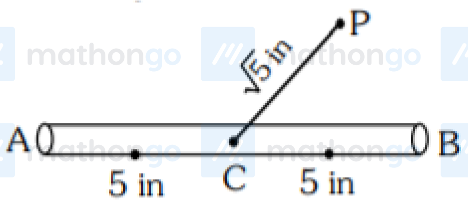
(2) $x^2 + 2(a+2b)x + a(a+b) = 0$

(3) $x^2 - 2ax + b(a+b) = 0$

(4) $x^2 - 2ax + a(a+b) = 0$

Q3 2021 (26 Aug Shift 2)

A 10 inches long pencil AB with mid point C and a small eraser P are placed on the horizontal top of a table such that $\mathrm{PC} = \sqrt{5}$ inches and $\angle \mathrm{PCB} = \tan^{-1}(2)$. The acute angle through which the pencil must be rotated about C so that the perpendicular distance between eraser and pencil becomes exactly 1 inch is:



(1) $\tan^{-1}\left(\frac{3}{4}\right)$

(2) $\tan^{-1}(1)$

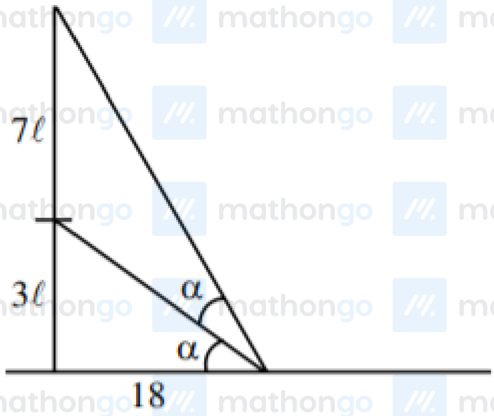
(3) $\tan^{-1}\left(\frac{4}{3}\right)$

(4) $\tan^{-1}\left(\frac{1}{2}\right)$

Answer Key

Q1 (2) **Q2** (3) **Q3** (1)

Q1 (2)



Let height of pole $= 10\ell$

$\tan\alpha = \frac{3\ell}{18} = \frac{\ell}{6}$

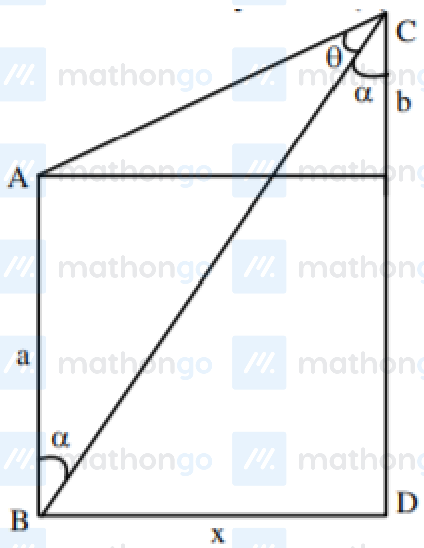
$\tan 2\alpha = \frac{10\ell}{18}$

$\frac{2\tan\alpha}{1-\tan^2\alpha} = \frac{10\ell}{18}$

use $\tan\alpha = \frac{\ell}{6} \Rightarrow \ell = \sqrt{\frac{72}{5}}$

height of pole $= 10\ell = 12\sqrt{10}$

Q2 (3)



$\tan\theta = \frac{1}{2}$

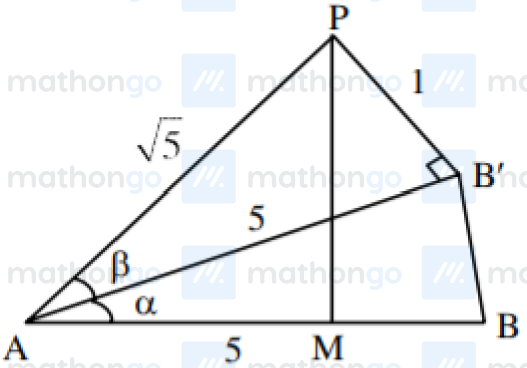
$\tan(\theta+\alpha)=\frac{\mathrm{x}}{\mathrm{b}}, \tan\alpha=\frac{\mathrm{x}}{\mathrm{a+b}}$

$\Rightarrow \frac{1}{2}+\frac{x}{a+b}$

$\Rightarrow \frac{\frac{1}{2}+\frac{x}{a+b}}{1-\frac{1}{2}\times\frac{x}{a+b}}=\frac{x}{b}.$

$\Rightarrow x^2-2ax+ab+b^2=0$

Q3 (1)



From figure.

$\sin\beta=\frac{1}{\sqrt{5}}$

$\therefore \tan\beta=\frac{1}{2}$

$\tan(\alpha+\beta)=2$

$\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\cdot\tan\beta}=2$

$\frac{\tan\alpha+\frac{1}{2}}{1-\tan\alpha\left(\frac{1}{2}\right)}=2$

$\tan\alpha=\frac{3}{4}$

$\alpha=\tan^{1}\left(\frac{3}{4}\right)$